

HOVICET FILM COATED TABLET 5 MG

VIHOVxx-0 [5]

DESCRIPTION

Oblong, white to off-white film-coated tablet, bevel-edged, shallow convex with 'HD' embossed on one face.

COMPOSITION

Each tablet contains 5 mg of Levocetirizine Dihydrochloride.

PHARMACODYNAMICS

Pharmacotherapeutic group
Antihistamine for systemic use, piperazine derivative.

ATC Code
R06A E09

Mechanism of Action/Pharmacodynamic Effects
Levocetirizine, the (R) enantiomer of cetirizine, is a potent and selective antagonist of peripheral H1-receptors.

Binding studies revealed that levocetirizine has high affinity for human H1-receptors (Ki = 3.2 nmol/L). Levocetirizine has an affinity 2-fold higher than that of cetirizine (Ki = 6.3 nmol/L). Levocetirizine dissociates from H1-receptors with a half-life of 115±38min.

After single administration, levocetirizine shows a receptor occupancy of 90% at 4 hours and 57% at 24 hours.

At half the dose, levocetirizine has comparable activity to cetirizine, both in the skin and in the nose.

Levocetirizine treatment resulted in significantly decreased wheal and flare formation which was highest in first 12 hours and lasted for 24 hours, compared with placebo and desloratadine.

The onset of action of levocetirizine has been observed at 1 hour post drug intake.

Levocetirizine shows to inhibit eotaxin-induced eosinophil transendothelial migration through blood dermal and lung cells. Three main inhibitory effect of levocetirizine 5mg in the first 6 hours of pollen-induced reaction, compared with placebo: inhibition of VCAM-1 release, modulation of vascular permeability and a decrease in eosinophil recruitment.

PHARMACOKINETICS

The pharmacokinetics of levocetirizine are linear with dose and time-independent with low inter-subject variability. The pharmacokinetic profile is the same when given as the single enantiomer or when given as cetirizine. No chiral inversion occurs during the process of absorption and elimination.

Absorption

Levocetirizine is rapidly and extensively absorbed following oral administration. In adults, peak plasma concentrations are achieved 0.9 h after dosing. Steady state is achieved after two days. Peak concentrations are typically 270 ng/ml and 308 ng/ml following a single and a repeated 5 mg once daily dose, respectively. The extent of absorption is dose-independent and is not altered by food, but the peak concentration is reduced and delayed.

Distribution

No tissue distribution data are available in humans, neither concerning the passage of levocetirizine through the blood-brain barrier. In rats and dogs, the highest tissue levels, are found in liver and kidneys, the lowest in the CNS compartment. In humans, levocetirizine is 90% bound to plasma proteins. The distribution of levocetirizine is restrictive, as the volume of distribution is 0.4 l/kg.

Metabolism

The extent of metabolism of levocetirizine in humans is less than 14% of the dose and therefore differences resulting from genetic polymorphism or concomitant intake of enzyme inhibitors are expected to be negligible. Metabolic pathways include aromatic oxidation, N- and O-dealkylation and taurine conjugation. Dealkylation pathways are primarily mediated by CYP 3A4 while aromatic oxidation involved multiple and/or unidentified CYP isoforms. Levocetirizine had no effect on the activities of CYP isoenzymes 1A2, 2C9, 2C19, 2D6, 2E1 and 3A4 at concentrations well above peak concentrations achieved following a 5 mg oral dose. Due to its low metabolism and absence of metabolic inhibition potential, the interaction of levocetirizine with other substances, or vice-versa, is unlikely.

Elimination

The plasma half-life in adults is 7.9 ± 1.9 hours. The half-life is shorter in small children. The mean apparent total body clearance in adults is 0.63 ml/min/kg. The major route of excretion of levocetirizine and metabolites is via urine, accounting for a mean of 85.4% of the dose. Excretion via faeces accounts for only 12.9% of the dose. Levocetirizine is excreted both by glomerular filtration and active tubular secretion.

Special patient populations

Children

Oral administration of a single dose of 5 mg levocetirizine in 14 children age 6 to 11 years with body weight ranging between 20 and 40 kg show that Cmax and AUC values are about 2-fold greater than that reported in healthy adult subjects in a cross-study comparison. The mean Cmax was 450 ng/mL, occurring at a mean time of 1.2 hours, weight-normalised, total body clearance was 30% greater, and the elimination half-life 24% shorter in this paediatric population than in adults.

Elderly

Limited pharmacokinetic data are available in elderly subjects. Following once daily repeat oral administration of 30 mg levocetirizine for 6 days, the total body clearance was approximately 33% lower compared to that in younger adults. The disposition of racemic cetirizine has been shown to be dependent on renal function rather than on age. This finding would also be applicable for levocetirizine, as levocetirizine and cetirizine are both predominantly excreted in urine. Therefore, the levocetirizine dose should be adjusted in accordance with renal function in elderly patients.

Renal impairment

The apparent body clearance of levocetirizine is correlated to the creatinine clearance. It is therefore recommended to adjust the dosing intervals of levocetirizine, based on creatinine clearance in patients with moderate and severe renal impairment. In anuric end stage renal disease subjects, the total body clearance is decreased by approximately 80% when compared to normal subjects. The amount of levocetirizine removed during a standard 4-hour hemodialysis procedure was < 10%.

Hepatic impairment

The pharmacokinetics of levocetirizine in hepatically impaired subjects have not been tested. Patients with chronic liver diseases (hepatocellular, cholestatic, and biliary cirrhosis) given 10 or 20 mg of the racemic compound cetirizine as a single dose had a 50% increase in half life along with a 40% decrease in clearance compared to healthy subjects.

Other patient characteristics

Gender

The half-life was slightly shorter in women (7.08 ± 1.72 hr) than in men (8.62 ± 1.84 hr); however, the body weight-adjusted oral clearance in women (0.67 ± 0.16 mL/min/kg) appears to be comparable to that in men (0.59 ± 0.12 mL/min/kg). The same daily doses and dosing intervals are applicable for men and women with normal renal function.

Race

As levocetirizine is primarily renally excreted, and there are no important racial differences in creatinine clearance, pharmacokinetic characteristics of levocetirizine are not expected to be different across races. No race-related differences in the kinetics of racemic cetirizine have been observed.

RECOMMENDED DOSE

Adults and adolescents 12 years and above

The daily recommended dose is 5mg (1 film-coated tablet).

Children

Children aged 6 to 12 years

The daily recommended dose is 5mg (1 film-coated tablet).

Elderly

Adjustment of the dose is recommended in elderly patients with moderate to severe renal impairment (see Renal impairment).

Renal impairment

The dosing intervals must be individualized according to renal function (eGFR – estimated Glomerular Filtration Rate). Refer to the following tablet and adjust the dose as indicated.

$$CL_{cr} = \frac{[140 - age \text{ (years)}] \times weight \text{ (kg)}}{72 \times serum \text{ creatinine (mg/dl)}} \times (0.85 \text{ for women})$$

Dosing Adjustments for Patients with Impaired Renal Function:

Group	eGFR (ml/min)	Dosage and frequency
Normal renal function	≥ 90	5 mg once daily
Mildly decreased renal function	60 – < 90	5 mg once daily
Moderately decreased renal function	30 – < 60	5 mg once daily once every 2 days
Severely decreased renal function	15 - < 30 (not requiring dialysis)	5 mg once daily once every 3 days
End stage renal disease (ESRD)	< 15 (requiring dialysis treatment)	Contraindicated

In paediatric patients suffering from renal impairment, the dose will have to be adjusted on an individual basis taking into account the renal clearance of the patient and his body weight. There are no specific data for children with renal impairment.

Hepatic impairment

No dose adjustment is needed in patients with solely hepatic impairment. In patients with hepatic impairment and renal impairment, adjustment of the dose is recommended (see Renal impairment).

INDICATION

Levocetirizine is indicated for the symptomatic treatment of allergic rhinitis (including persistent allergic rhinitis) and chronic idiopathic urticaria.

CONTRAINDICATIONS

Levocetirizine is contraindicated in:

- hypersensitivity to levocetirizine, to cetirizine, to hydroxyzine, to any piperazine derivatives or to any of the excipients,

- patient with end stage renal disease with estimated Glomerular Filtration Rate (eGFR) below 15 ml/min (requiring dialysis treatment).

WARNINGS AND PRECAUTIONS

Alcohol
Precaution is recommended with concurrent intake of alcohol (see Section Interactions with other medicaments).

Risk of urinary retention.
Caution should be taken in patients with predisposing factors of urinary retention (e.g. spinal cord lesion, prostatic hyperplasia) as levocetirizine may increase the risk of urinary retention.

Risk of seizure aggravation
Caution should be taken in patients with epilepsy and patients at risk of convulsion as levocetirizine may cause seizure aggravation.

Allergy skin tests.
Response to allergy skin tests are inhibited by antihistamines and a wash-out period (of 3 days) is required before performing them.

Withdrawal syndrome.
Pruritus may occur when levocetirizine is stopped even if those symptoms were not present before treatment initiation (see Section Adverse Reactions). The symptoms may resolve spontaneously. In some cases, the symptoms may be intense and may require treatment to be restarted. The symptoms should resolve when the treatment is restarted.

Excipients.
Lactose
This medicinal product contains lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

INTERACTIONS WITH OTHER MEDICAMENTS

No interaction studies have been performed with levocetirizine (including no studies with CYP3A4 inducers); studies with the racemate compound cetirizine demonstrated that there were no clinically relevant adverse interactions (with antipyrine, pseudoephedrine, cimetidine, ketoconazole, erythromycin, azithromycin, glipizide and diazepam).

Theophylline
A small decrease in the clearance of cetirizine (16%) was observed in a multiple dose study with theophylline (400 mg once a day); while the disposition of theophylline was not altered by concomitant cetirizine administration.

Ritonavir
In a multiple dose study of ritonavir (600 mg twice daily) and cetirizine (10 mg daily), the extent of exposure to cetirizine was increased by about 40% while the disposition of ritonavir was slightly altered (-11%) further to concomitant cetirizine administration.

Food
The extent of absorption of levocetirizine is not reduced with food, although the rate of absorption is decreased.

Alcohol
In sensitive patients the concurrent administration of cetirizine or levocetirizine and alcohol or other CNS depressants may cause additional reductions in alertness and impairment of performance.

PREGNANCY AND LACTATION

Fertility
There are no relevant data available.

Pregnancy
The use of levocetirizine may be considered during pregnancy, if necessary. There are no or limited amount of data (less than 300 pregnancy outcomes) from the use of levocetirizine in pregnant women. However, for cetirizine, the racemate of levocetirizine, a large amount of data (more than 1000 pregnancy outcomes) on pregnant women indicate no malformative or feto/neonatal toxicity. Animal studies do not indicate direct or indirect harmful effects with respect to pregnancy, embryonal/fetal development, parturition or postnatal development.

Lactation
Caution should be exercised when prescribing to lactating women. Cetirizine, the racemate of levocetirizine has been shown excreted in human. Therefore, the excretion of levocetirizine in human milk is likely. Adverse reactions associated with levocetirizine may be observed in breastfed infants.

SIDE EFFECTS

Adverse drug reactions (ADRs) are listed below by MedDRA system organ class and by frequency.

MedDRA System Organ Class	Adverse Reaction	Frequency
Nervous system disorders	Headache, somnolence	Common
	Convulsions, paraesthesia, dizziness, syncope, tremor, dysgeusia	Not known
Gastrointestinal disorders	Dry mouth	Common
	Abdominal pain	Uncommon
	Nausea, vomiting, diarrhoea	Not known
General disorders and administration site conditions	Fatigue	Common
	Asthenia	Uncommon
	Oedema	Not known

Hovicet Film Coated Tablet 5 Mg

Immune system disorders	Hypersensitivity including anaphylaxis	Not known
Metabolism and nutrition disorders	Increase weight, increased appetite	Not known
Psychiatric disorders	Aggression, agitation, hallucination, depression, insomnia, suicidal ideation, nightmares	Not known
Eye disorders	Visual disturbances, blurred vision, oculogyration	Not known
Ear and labyrinth disorders	Vertigo	Not known
Cardiac disorders	Palpitation, tachcardia	Not known
Respiratory, thoracic, and mediastinal disorders	Dyspnoea	Not known
Hepatobiliary disorders	Hepatitis, abnormal liver function test	Not known
Skin and subcutaneous tissue disorders	Angioneurotic oedema, fixed drug eruption, pruritus, rash, urticaria	Not known
Musculoskeletal and connective tissue disorders	Myalgia, arthralgia	Not known
Renal and urinary disorders	Dysuria, urinary retention	Not known

Skin reactions occurring after discontinuation of levocetirizine
After levocetirizine discontinuation, pruritus has been reported (see Section Warnings and Precautions).

SYMPTOMS AND TREATMENT OF OVERDOSE

Symptoms and Signs
Symptoms of overdose may include drowsiness in adults. In children, agitation and restlessness may initially occur, followed by drowsiness.

Treatment
There is no known specific antidote to levocetirizine. Should overdose occur, symptomatic or supportive treatment is recommended. Levocetirizine is not effectively removed by haemodialysis.

Further management should be as clinically indicated or as recommended by the national poisons centre, where available.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

Ability to perform tasks that require judgement, motor or cognitive skills
No evidence that levocetirizine at the recommended dose impairs mental alertness, reactivity or the ability to drive and use machines.

Nevertheless, some patients could experience somnolence, fatigue and asthenia under therapy with levocetirizine. Therefore, patients intending to drive, engage in potentially hazardous activities or operate machinery should take their response to the medicinal product into account.

Storage: Store below 30°C.

Presentation/ Packing: Alu/Alu Blister of 10 x 10's

Product registration holder (Malaysia) /
Product owner: HOVID Berhad
121, Jalan Tunku Abdul Rahman (Jalan Kuala Kangsar),
30010 Ipoh, Perak, Malaysia.

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